ROHIT BANKAR

(+1) 765-767-2732 | rohitbankar2000@gmail.com | LinkedIn | GitHub | Medium | Portfolio

EDUCATION

Purdue University *January'24 - December'25*
Master of Science in Computer and Information Technology ***(4.0/4.0)*** *West Lafayette, IN*

Indian Institute of Technology (IIT) Indore *July'18 - May'22*
Bachelor's in Technology (BTech) **(8.77/10.0)**

PROFESSIONAL EXPERIENCE

Amazon (AWS Bedrock) *May'25- August'25*
Software Development Intern *Seattle, USA*

- Developed **Hubble**, a centralized service for managing **Anthropic Claude LLM** models (Opus, Sonnet, Haiku) on AWS Bedrock, integrating with Trainium-based infrastructure to optimize deployments and reduce manual intervention by 90%.
- Enhanced the Forklift team's operational efficiency by optimizing **inference latency, reducing throttling, and integrating API-based capacity management with automated rebalance planning**, reducing issue-resolution time by 50%**.**
- Implemented and integrated **15+ RESTful APIs,** enhancing visibility and management efficiency for Claude replicas.

Razorpay (South Asia's Largest Payment Gateway) *July'22- December'23*
Software Development Engineer (Backend) *Bengaluru, India*

- Interacted with 5+ teams to lead the re-architecture of Harvester, a real-time Spark-based data service, **saving $10-11k /month.**
- Coded custom whitelisting logic for Maxwell and Debezium CDC collectors, securing ~90% of company data.
- Migrated SRM dashboard **from TiDb to Pinot, reducing lag from 5 to 1 minute**, enhancing experience of 5+ teams & clients.
- Integrated Schema Registry into entity onboarding flows and ETL pipelines by implementing necessary transformation logic and handling horizontal and vertical scaling of pods, resulting in better versioning and 100% faster retrieval of schemas.
- Improved on-call bot's data flow, adding features for real-time updates, achieving 40% faster resolution.

Jaguar Land Rover *June'21- August'21*
Software Development Intern *Bengaluru, India*

- Designed and coded a Machine Learning Pipeline to extract critical information from handwritten sentences using standard NLP libraries **like Stanford-openie and Textacy, achieving 70%** improvement over previous algorithms.
- Made exponential smoothing and **LSTM-RNN models, boosting failure prediction accuracy from 80% to over 90%.**
- Wrote a model to quantify optically scanned data using string-matching algorithms like Levenshtein-Damerau and Jaro-Winkler, achieving desired results with 95% accuracy.

Purdue University *January'24 - Present*
Software Development Engineer (GRA) *West Lafayette, USA*

- Built 8+ critical data pipelines to ingest thousands of daily data points from Qualtrics & Salesforce to on-premise databases.
- Created and maintained **10+ data pipelines** for secure, efficient data transfer. Generated data visualizations for 20+ clients.
- Develop, host, and maintain a scalable website for the NSF-funded project Advancing Cybersecurity Education.

RESEARCH & PROJECTS

Compute and Memory resource estimation for ML pipelines *January'24 – December'25*
MS Research - Thesis Project
Publications: IEEE ANTS 2025, IEEE ICC 2026

- Designed a predictive metamodel for machine learning models' CPU/GPU & memory requirements, reducing cost by 25%.
- Proposed and validated **CORE-ML**, a stage-aware resource allocation framework for non-inference pipelines, reducing CPU over-provisioning by **up to 80%** and overall costs by **up to 50%**.
- Built **DeepScale**, a real-time Kubernetes VPA–based scaling system for training pods, cutting compute costs by **60%** and CPU-hours by **up to 85%** via high-frequency in-place scaling.

Detection of Suicidal Ideation through Deep Learning *March'23 - October'23*
Research Project - IIT Indore

- Developed a model using BERT, Graph Sage, and Hierarchical Attention to classify suicidal texts with **an F1 score > 0.975.**
- Enhanced model accuracy by integrating lexicons and emotional analysis, achieving a total F1 score increase of 6-7%.

Open-Source Contributions *January'21 – December'23*

- Programmed a Go authentication module for **Trino gateway**, enhancing security & access control for 2000+ Razorpay users.
- Integrated credential authentication into Query Book's UI, benefiting 10000+ users.
- Engineered a tool to convert LookML to Cube.js at Razorpay's annual hackathon, automating the process and reducing resource consumption by at **least 70%; received organization-wide recognition and bounty.**

SKILLS

- **Languages:** Python, Golang, Scala, C++, SQL, Java, JavaScript, TypeScript, R, Rust, HTML, CSS, MATLAB, Node.
- **Tools:** Databricks, AWS, GCP, Docker, MongoDB, Redis, ELT, GitHub, Kafka, Git, MySQL, PostgreSQL, Linux, DynamoDB.
- **Technologies:** Spark, React, NextJS, SDLC, Airflow, Django, PyTorch, Kubernetes, Flask, TensorFlow, CI/CD, Jenkins.
- **Extra-curricular/Awards:** Head Chess Club IITI, Exceptional Work Award for HMS Upgrade, Inter-IIT Sports Meet.